Contents

1 NAME OF EXPERIMENT **2**

2 OBJECTIVE **2**

3 THEORY **2**

3.1 Equivalence Class Testing 2
3.2 Types of Equivalence Classes 2
3.3 Testing Approaches 2
3.4 System Under Test: Student Registration Validator 3

4 TEST CASES **3**

4.1 Equivalence Classes Definition 3
4.2 Generated Test Cases 5
4.2.1 Weak Normal Equivalence Class Test Cases 5
4.2.2 Weak Robust Equivalence Class Test Cases 5
4.2.3 Strong Normal Equivalence Class Test Cases 6
4.2.4 Strong Robust Equivalence Class Test Cases 6

5 SCREENSHOTS **8**

6 BUG IDENTIFICATION **10**

6.1 Testing Summary 10
6.2 Bugs Identified 10
6.3 Potential Improvements 10

7 CONCLUSION **11**

1 NAME OF EXPERIMENT

Equivalence Class Testing of Student Registration Validation System

2 OBJECTIVE

The objective of this lab exercise is to:

- Understand and apply equivalence class testing technique for software validation
- Design and identify valid and invalid equivalence classes for the Student Registration Validation System
- Generate comprehensive test cases using Weak Normal, Weak Robust, Strong Normal, and Strong Robust testing approaches
- Execute test cases and verify the actual output against expected output
- Identify bugs and defects in the validation system through systematic testing

3 THEORY

3.1 Equivalence Class Testing

Equivalence Class Testing (ECT) is a black-box testing technique that divides the input domain of a program into classes of data from which test cases can be derived. The fundamental principle is that test cases designed from an equivalence class are equally effective in uncovering defects, so testing one representative value from each class is sufficient.

Each equivalence class represents a set of valid or invalid states for system inputs. The key assumption is that if one test case in an equivalence class detects a defect, all other test cases in that class would likely detect the same defect. Conversely, if one test case does not detect a defect, none of the others in that class will either.

3.2 Types of Equivalence Classes

Valid Equivalence Classes: These represent valid input conditions that the system should accept and process correctly. For the Student Registration System, valid classes include ages between 16-25, CGPA between 2.0-4.0, and credits between 0-150.

Invalid Equivalence Classes: These represent invalid input conditions that should be rejected by the system. Examples include ages below 16 or above 25, CGPA below 2.0 or above 4.0, and negative credits or credits exceeding 150.

3.3 Testing Approaches

Weak Normal Equivalence Class Testing: Tests one variable from each valid equivalence class while keeping others in their valid ranges. Assumes that errors are independent of each other. Minimum number of test cases equals the number of equivalence classes.

Weak Robust Equivalence Class Testing: Extends Weak Normal by including one test case from each invalid equivalence class, with all other variables in valid ranges. Tests boundary violations independently.

Strong Normal Equivalence Class Testing: Tests all possible combinations of one value from each valid equivalence class. Assumes interactions between variables may reveal defects. Number of test cases equals the product of valid equivalence classes for each variable.

Strong Robust Equivalence Class Testing: The most comprehensive approach, testing all combinations of valid and invalid equivalence classes. Provides maximum coverage but requires the most test cases.

3.4 System Under Test: Student Registration Validator

The Student Registration Validation System validates three input parameters before allowing student registration. Each parameter has specific validation rules:

Input Specifications:

Parameter	Valid Range	Type	Error Messages
Age	16 to 25 (inclusive)	Integer	< 16: "Age too young for registration" > 25: "Age exceeds maximum limit"
CGPA	2.0 to 4.0 (inclusive)	Float	< 2.0: "CGPA below minimum requirement" > 4.0: "Invalid CGPA value"
Credits	0 to 150 (inclusive)	Integer	< 0: "Credits cannot be negative" > 150: "Credits exceed maximum allowed"

Table 1: Input Specifications for Student Registration Validator

4 TEST CASES

4.1 Equivalence Classes Definition

Age Equivalence Classes:

Class ID	Type	Description
EC1	Invalid	Age < 16 (Too young)
EC2	Valid	Age between 16 and 25 (inclusive)
EC3	Invalid	Age > 25 (Too old)

Table 2: Age Equivalence Classes

CGPA Equivalence Classes:

Class ID	Type	Description
EC4	Invalid	CGPA < 2.0 (Below minimum)
EC5	Valid	CGPA between 2.0 and 4.0 (inclusive)
EC6	Invalid	CGPA > 4.0 (Above maximum)

Table 3: CGPA Equivalence Classes

Credits Equivalence Classes:

Class ID	Type	Description
EC7	Invalid	Credits < 0 (Negative)
EC8	Valid	Credits between 0 and 150 (inclusive)
EC9	Invalid	Credits > 150 (Exceeds maximum)

Table 4: Credits Equivalence Classes

4.2 Generated Test Cases

4.2.1 Weak Normal Equivalence Class Test Cases

These test cases use one value from each valid equivalence class with all inputs in valid ranges.

Test ID	Age	CGPA	Credits	Equivalence Classes	Expected Output	Status
TC001	20	3.0	75	EC2, EC5, EC8	Registration Approved	PASS

Table 5: Weak Normal Test Cases

4.2.2 Weak Robust Equivalence Class Test Cases

These test cases include invalid values for each parameter while keeping others valid.

Test ID	Age	CGPA	Credits	Equiv. Classes	Expected Output	Status
TC002	15	3.0	75	EC1, EC5, EC8	Age too young for registration	PASS
TC003	26	3.0	75	EC3, EC5, EC8	Age exceeds maximum limit	PASS
TC004	20	1.5	75	EC2, EC4, EC8	CGPA below minimum requirement	PASS
TC005	20	4.5	75	EC2, EC6, EC8	Invalid CGPA value	PASS
TC006	20	3.0	-10	EC2, EC5, EC7	Credits cannot be negative	PASS
TC007	20	3.0	160	EC2, EC5, EC9	Credits exceed maximum allowed	PASS

Table 6: Weak Robust Test Cases

4.2.3 Strong Normal Equivalence Class Test Cases

These test cases test various combinations of valid values from all equivalence classes.

Test ID	Age	CGPA	Credits	Equiv. Classes	Expected Output	Status
TC008	16	2.0	0	EC2, EC5, EC8	Registration Approved	PASS
TC009	16	2.0	150	EC2, EC5, EC8	Registration Approved	PASS
TC010	16	4.0	0	EC2, EC5, EC8	Registration Approved	PASS
TC011	16	4.0	150	EC2, EC5, EC8	Registration Approved	PASS
TC012	25	2.0	0	EC2, EC5, EC8	Registration Approved	PASS
TC013	25	2.0	150	EC2, EC5, EC8	Registration Approved	PASS
TC014	25	4.0	0	EC2, EC5, EC8	Registration Approved	PASS
TC015	25	4.0	150	EC2, EC5, EC8	Registration Approved	PASS

Table 7: Strong Normal Test Cases

4.2.4 Strong Robust Equivalence Class Test Cases

These test cases test ALL combinations of valid and invalid equivalence classes

Test ID	Age	CGPA	Credits	Equiv. Classes	Expected Output	Status
TC016	15	1.5	-10	EC1, EC4, EC7	Age too young for registration	PASS
TC017	15	1.5	75	EC1, EC4, EC8	Age too young for registration	PASS
TC018	15	1.5	160	EC1, EC4, EC9	Age too young for registration	PASS
TC019	15	3.0	-10	EC1, EC5, EC7	Age too young for registration	PASS
TC020	15	3.0	75	EC1, EC5, EC8	Age too young for registration	PASS
TC021	15	3.0	160	EC1, EC5, EC9	Age too young for registration	PASS
TC022	15	4.5	-10	EC1, EC6, EC7	Age too young for registration	PASS
TC023	15	4.5	75	EC1, EC6, EC8	Age too young for registration	PASS

Table 8 –

Test ID	Age	CGPA	Credits	Equiv. Classes	Expected Output	Status
TC024	15	4.5	160	EC1, EC6, EC9	Age too young for registration	PASS
TC025	20	1.5	-10	EC2, EC4, EC7	CGPA below minimum requirement	PASS
TC026	20	1.5	75	EC2, EC4, EC8	CGPA below minimum requirement	PASS
TC027	20	1.5	160	EC2, EC4, EC9	CGPA below minimum requirement	PASS
TC028	20	3.0	-10	EC2, EC5, EC7	Credits cannot be negative	PASS
TC029	20	3.0	75	EC2, EC5, EC8	Registration Approved	PASS
TC030	20	3.0	160	EC2, EC5, EC9	Credits exceed maximum allowed	PASS
TC031	20	4.5	-10	EC2, EC6, EC7	Invalid CGPA value	PASS
TC032	20	4.5	75	EC2, EC6, EC8	Invalid CGPA value	PASS
TC033	20	4.5	160	EC2, EC6, EC9	Invalid CGPA value	PASS
TC034	26	1.5	-10	EC3, EC4, EC7	Age exceeds maximum limit	PASS
TC035	26	1.5	75	EC3, EC4, EC8	Age exceeds maximum limit	PASS
TC036	26	1.5	160	EC3, EC4, EC9	Age exceeds maximum limit	PASS
TC037	26	3.0	-10	EC3, EC5, EC7	Age exceeds maximum limit	PASS
TC038	26	3.0	75	EC3, EC5, EC8	Age exceeds maximum limit	PASS
TC039	26	3.0	160	EC3, EC5, EC9	Age exceeds maximum limit	PASS
TC040	26	4.5	-10	EC3, EC6, EC7	Age exceeds maximum limit	PASS
TC041	26	4.5	75	EC3, EC6, EC8	Age exceeds maximum limit	PASS
TC042	26	4.5	160	EC3, EC6, EC9	Age exceeds maximum limit	PASS

Table 8: Strong Robust Equivalence Class Test Cases

5 SCREENSHOTS

Screenshots showing the tabulated equivalence class test cases and their results for the Student Registration Validation System, documented in an Excel worksheet.

		WEAK NORMAL & WEAK ROBUST TEST CASES		
Test ID	**Inputs**	**Expected Output**	**Actual Output**	**Status**
TC001	Age=20, CGPA=3.0, Credits=75	Registration Approved	Registration Approved	PASS
TC002	Age=15, CGPA=3.0, Credits=75	Age too young for registration	Age too young for registration	PASS
TC003	Age=26, CGPA=3.0, Credits=75	Age exceeds maximum limit	Age exceeds maximum limit	PASS
TC004	Age=20, CGPA=1.5, Credits=75	CGPA below minimum requiremen	CGPA below minimum requiremen	PASS
TC005	Age=20, CGPA=4.5, Credits=75	Invalid CGPA value	Invalid CGPA value	PASS
TC006	Age=20, CGPA=3.0, Credits=-10	Credits cannot be negative	Credits cannot be negative	PASS
TC007	Age=20, CGPA=3.0, Credits=160	Credits exceed maximum allowed	Credits exceed maximum allowed	PASS

Figure 1: Execution of Weak Normal and Weak Robust Equivalence Class Test Cases

		STRONG NORMAL TEST CASES		
Test ID	**Inputs**	**Expected Output**	**Actual Output**	**Status**
TC008	Age=16, CGPA=2.0, Credits=0	Registration Approved	Registration Approved	PASS
TC009	Age=16, CGPA=2.0, Credits=150	Registration Approved	Registration Approved	PASS
TC010	Age=16, CGPA=4.0, Credits=0	Registration Approved	Registration Approved	PASS
TC011	Age=16, CGPA=4.0, Credits=150	Registration Approved	Registration Approved	PASS
TC012	Age=25, CGPA=2.0, Credits=0	Registration Approved	Registration Approved	PASS
TC013	Age=25, CGPA=2.0, Credits=150	Registration Approved	Registration Approved	PASS
TC014	Age=25, CGPA=4.0, Credits=0	Registration Approved	Registration Approved	PASS
TC015	Age=25, CGPA=4.0, Credits=150	Registration Approved	Registration Approved	PASS

Figure 2: Execution of Strong Normal Equivalence Class Test Cases

		STRONG ROBUST TEST CASES		
Test ID	**Inputs**	**Expected Output**	**Actual Output**	**Status**
TC016	Age=15, CGPA=1.5, Credits=-10	Age too young for registration	Age too young for registration	PASS
TC017	Age=15, CGPA=1.5, Credits=75	Age too young for registration	Age too young for registration	PASS
TC018	Age=15, CGPA=1.5, Credits=160	Age too young for registration	Age too young for registration	PASS
TC019	Age=15, CGPA=3.0, Credits=-10	Age too young for registration	Age too young for registration	PASS
TC020	Age=15, CGPA=3.0, Credits=75	Age too young for registration	Age too young for registration	PASS
TC021	Age=15, CGPA=3.0, Credits=160	Age too young for registration	Age too young for registration	PASS
TC022	Age=15, CGPA=4.5, Credits=-10	Age too young for registration	Age too young for registration	PASS
TC023	Age=15, CGPA=4.5, Credits=75	Age too young for registration	Age too young for registration	PASS
TC024	Age=15, CGPA=4.5, Credits=160	Age too young for registration	Age too young for registration	PASS
TC025	Age=20, CGPA=1.5, Credits=-10	CGPA below minimum requirement	CGPA below minimum requirement	PASS
TC026	Age=20, CGPA=1.5, Credits=75	CGPA below minimum requirement	CGPA below minimum requirement	PASS
TC027	Age=20, CGPA=1.5, Credits=160	CGPA below minimum requirement	CGPA below minimum requirement	PASS
TC028	Age=20, CGPA=3.0, Credits=-10	Credits cannot be negative	Credits cannot be negative	PASS
TC029	Age=20, CGPA=3.0, Credits=75	Registration Approved	Registration Approved	PASS
TC030	Age=20, CGPA=3.0, Credits=160	Credits exceed maximum allowed	Credits exceed maximum allowed	PASS
TC031	Age=20, CGPA=4.5, Credits=-10	Invalid CGPA value	Invalid CGPA value	PASS
TC032	Age=20, CGPA=4.5, Credits=75	Invalid CGPA value	Invalid CGPA value	PASS
TC033	Age=20, CGPA=4.5, Credits=160	Invalid CGPA value	Invalid CGPA value	PASS
TC034	Age=26, CGPA=1.5, Credits=-10	Age exceeds maximum limit	Age exceeds maximum limit	PASS
TC035	Age=26, CGPA=1.5, Credits=75	Age exceeds maximum limit	Age exceeds maximum limit	PASS
TC036	Age=26, CGPA=1.5, Credits=160	Age exceeds maximum limit	Age exceeds maximum limit	PASS
TC037	Age=26, CGPA=3.0, Credits=-10	Age exceeds maximum limit	Age exceeds maximum limit	PASS
TC038	Age=26, CGPA=3.0, Credits=75	Age exceeds maximum limit	Age exceeds maximum limit	PASS
TC039	Age=26, CGPA=3.0, Credits=160	Age exceeds maximum limit	Age exceeds maximum limit	PASS
TC040	Age=26, CGPA=4.5, Credits=-10	Age exceeds maximum limit	Age exceeds maximum limit	PASS
TC041	Age=26, CGPA=4.5, Credits=75	Age exceeds maximum limit	Age exceeds maximum limit	PASS
TC042	Age=26, CGPA=4.5, Credits=160	Age exceeds maximum limit	Age exceeds maximum limit	PASS

Figure 3: Execution of Strong Robust Equivalence Class Test Cases

6 BUG IDENTIFICATION

6.1 Testing Summary

Metric	Value
Total Test Cases Executed	42
Weak Normal Test Cases	1
Weak Robust Test Cases	6
Strong Normal Test Cases	8
Strong Robust Test Cases	27
Test Cases Passed	42
Test Cases Failed	0
Pass Rate	100%

Table 9: Test Execution Summary

6.2 Bugs Identified

Based on the equivalence class testing performed using Weak Normal, Weak Robust, Strong Normal, and Strong Robust testing strategies on the Student Registration Validation System, the following observations were made:

Observation 1: All test cases executed successfully, and the actual outputs matched the expected outputs for all equivalence classes.

Observation 2: The system correctly validates age, CGPA, and credit constraints and handles invalid input values appropriately.

Observation 3: Boundary and combination testing confirmed correct behavior at minimum and maximum limits for all input parameters.

Observation 4: The error messages generated by the system are clear, specific, and accurately describe the reason for validation failure.

6.3 Potential Improvements

- Introduce validation for decimal precision in CGPA values.
- Add test cases to validate incorrect data types (e.g., age as floating-point values).
- Include test cases for non-numeric inputs to further improve robustness.
- Consider stress testing with extremely large values to evaluate system stability.

7 CONCLUSION

This lab exercise successfully demonstrated the application of equivalence class testing techniques to validate the Student Registration Validation System. The input domain was systematically partitioned into equivalence classes, and test cases were designed using Weak Normal, Weak Robust, Strong Normal, and Strong Robust testing strategies.

A total of 42 test cases were designed and executed, covering all valid and invalid equivalence classes as well as their possible combinations for the three input parameters: Age, CGPA, and Credits. All test cases passed successfully, with actual outputs matching the expected outputs, confirming that the system behaves correctly according to its specified validation rules.

The application of equivalence class testing proved effective in:

- Systematically organizing test cases based on valid and invalid input domains
- Ensuring comprehensive coverage of boundary values and input combinations
- Identifying potential error conditions through robust testing strategies
- Verifying the correctness and reliability of input validation logic

This experiment highlights the importance of structured software testing methodologies in quality assurance. By applying equivalence class testing, the system was validated efficiently while maintaining thorough coverage across all possible scenarios.

The knowledge and practical experience gained from this lab exercise will be valuable in real-world software testing, where equivalence class testing is widely used to improve test effectiveness, reduce redundancy, and ensure software reliability.